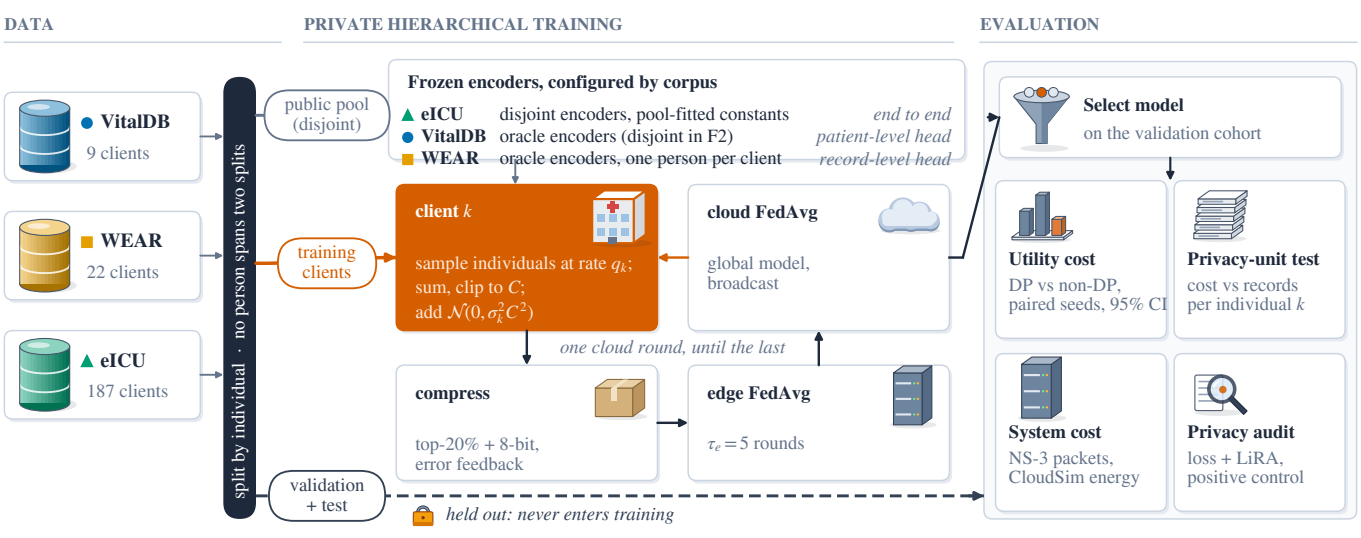
DATA
PRIVATE HIERARCHICAL TRAINING
EVALUATION
VitalDB
9 clients
WEAR
22 clients
eICU
187 clients
split by individual · no person spans two splits
public pool (disjoint)
training clients
validation + test
Frozen encoders, configured by corpus
eICU disjoint encoders, pool-fitted constants
VitalDB oracle encoders (disjoint in F2)
WEAR oracle encoders, one person per client
end to end
patient-level head
record-level head
client k
sample individuals at rate q_k; sum, clip to C; add $\mathcal{N}(0, \sigma_k^2 C^2)$
cloud FedAvg
global model, broadcast
one cloud round, until the last
compress
top-20% + 8-bit, error feedback
edge FedAvg
$\tau_e = 5$ rounds
held out: never enters training
Select model
on the validation cohort
Utility cost
DP vs non-DP, paired seeds, 95% CI
Privacy-unit test
cost vs records per individual k
System cost
NS-3 packets, CloudSim energy
Privacy audit
loss + LiRA, positive control